3. $\int (2+\tan x)^2 dx = \int 4 + 4\tan x + \tan^2 x \, dx$

$= \int 4 + 4\tan x + \sec^2 x - 1 \, dx$

$= \int 3 + 4\tan x + \sec^2 x \, dx$

$= \boxed{3x + 4\ln|\sec x| + \tan x + C}$

4. $\int_{\pi/6}^{\pi/2} x\csc^2 x \, dx = -x\cot x + \int_{\pi/6}^{\pi/2} \cot x \, dx$

$\boxed{\begin{array}{ll} u = x & dv = \csc^2 x \, dx \\ du = dx & v = -\cot x \end{array}}$ $= -x\cot x + \ln|\sin x| \Big]_{\pi/6}^{\pi/2}$

$uv - \int v\,du$

$= \left[-\frac{\pi}{2}\underbrace{\cot\left(\frac{\pi}{2}\right)}_{0} + \underbrace{\ln\left(\sin\frac{\pi}{2}\right)}_{0}\right] - \left[-\frac{\pi}{6}\underbrace{\cot\left(\frac{\pi}{6}\right)}_{\sqrt{3}} + \ln\left(\underbrace{\sin\frac{\pi}{6}}_{\frac{1}{2}}\right)\right]$

$= \boxed{\frac{\pi}{6}\cdot\sqrt{3} - \ln(1/2)}$

7. $\int_0^1 x^2\sqrt{2-x^2}\,dx = \int 2\sin^2\theta \, \sqrt{2}\cos\theta \, \sqrt{2}\cos\theta \, d\theta$

$a^2 - u^2$

$a = \sqrt{2} \quad u = x$

$\boxed{\begin{array}{l} x = \sqrt{2}\sin\theta \\ dx = \sqrt{2}\cos\theta \, d\theta \\ \sqrt{2-(\sqrt{2}\sin\theta)^2} = \sqrt{2-2\sin^2\theta} \\ = \sqrt{2(1-\sin^2\theta)} = \sqrt{2\cos^2\theta} \\ = \sqrt{2}\cos\theta \end{array}}$

$\int 2\sin^2\theta \cdot 2\cos^2\theta \, d\theta$

$= \int 2\cdot\frac{1}{2}\left[1-\cos 2\theta\right]\cdot 2\cdot\frac{1}{2}\left[1+\cos 2\theta\right] d\theta$

$= \int \left[1-\cos 2\theta\right]\left[1+\cos 2\theta\right] d\theta$

$= \int 1 - \cos^2 2\theta \, d\theta = \int 1 - \frac{1}{2}\left[1+\cos(4\theta)\right] d\theta$

Top: $1 = \sqrt{2}\sin\theta$ $\quad \theta = \frac{\pi}{4}$

$\frac{1}{\sqrt{2}} = \sin\theta$

Bot: $0 = \sqrt{2}\sin\theta \quad \theta = 0$

$$= \int 1 - \frac{1}{2} - \frac{1}{2}\cos 4\theta \, d\theta = \int \frac{1}{2} - \frac{1}{2}\cos 4\theta \, d\theta$$

$$= \frac{1}{2}\theta - \frac{1}{2}\frac{1}{4}\sin(4\theta)\Big]_0^{\pi/4} = \left(\frac{1}{2}\cdot\frac{\pi}{4} - \frac{1}{2}\cdot\frac{1}{4}\sin\pi\right) - \left(0 - \underbrace{\frac{1}{8}\sin 0}_{0}\right)$$

$$= \boxed{\frac{\pi}{8}}$$

8. $\int \frac{6x^2 - 3x + 1}{(4x+1)(x^2+1)}\,dx$

$$\left[\frac{6x^2 - 3x + 1}{(4x+1)(x^2+1)} = \frac{A}{4x+1} + \frac{Bx+C}{x^2+1}\right](4x+1)(x^2+1)$$

$$6x^2 - 3x + 1 = A(x^2+1) + (Bx+C)(4x+1)$$

$x = -\frac{1}{4}$

$$6\cdot\frac{1}{16} - 3\left(-\frac{1}{4}\right) + 1 = A\left(\frac{1}{16}+1\right) + 0$$

$$\frac{17}{8} = \frac{17}{16}A$$

$$2 = \frac{17}{8}\cdot\frac{16}{17} = A$$

$$6x^2 - 3x + 1 = 2x^2 + 2 + 4Bx^2 + Bx + 4Cx + C$$

x^2

$6 = 2 + 4B$

$1 = B$

const

$1 = 2 + C$

$-1 = C$

$$\int \frac{2}{4x+1} + \frac{x-1}{x^2+1}\,dx = \int \frac{2}{4x+1}\,dx + \int \frac{x}{x^2+1}\,dx - \int \frac{1}{x^2+1}\,dx$$

ln u-sub arctan

$u = x^2+1$

$du = 2x\,dx$

$$\int \frac{x}{u}\cdot\frac{du}{2x}$$

$$= \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln u$$

$$= \boxed{2\cdot\frac{1}{4}\ln|4x+1| + \frac{1}{2}\ln(x^2+1) - \arctan x + C}$$

Similar Example

$$\int \frac{3x+4}{x^2+9}\,dx = \int \frac{3x}{x^2+9}\,dx + \int \frac{4}{x^2+9}\,dx$$

u-sub arctan

13. $\int \frac{x^2+4x-45}{x^3}\,dx = \int \frac{x^2}{x^3} + \frac{4x}{x^3} - \frac{45}{x^3}\,dx$

$= \int \frac{1}{x} + \frac{4}{x^2} - \frac{45}{x^3}\,dx = \int \frac{1}{x} + 4x^{-2} - 45x^{-3}\,dx$

$= \ln|x| + \frac{4x^{-1}}{-1} - \frac{45x^{-2}}{-2} + C = \boxed{\ln|x| - \frac{4}{x} + \frac{45}{2x^2} + C}$

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5. $\int e^{2x}\sin x\,dx = -e^{2x}\cos x - \int 2e^{2x}(-\cos x)\,dx$

$\boxed{\begin{array}{ll} u = e^{2x} & dv = \sin x\,dx \\ du = 2e^{2x}dx & v = -\cos x \end{array}}$

$= -e^{2x}\cos x + 2\int e^{2x}\cos x\,dx$

$\boxed{\begin{array}{ll} u = e^{2x} & dv = \cos x\,dx \\ du = 2e^{2x}dx & v = \sin x \end{array}}$

$= -e^{2x}\cos x + 2\left[e^{2x}\sin x - \int 2e^{2x}\sin x\,dx\right]$

$I = -e^{2x}\cos x + 2e^{2x}\sin x - 4\underbrace{\int e^{2x}\sin x\,dx}_{I}$

$+4I \qquad\qquad +4I$

$\frac{5I}{5} = \boxed{\frac{1}{5}\left(-e^{2x}\cos x + 2e^{2x}\sin x\right) + C}$

Tabular:

$\int (\text{polynomial}) \cdot \left\{ \begin{array}{c} \text{sine} \\ \text{cosine} \\ e \end{array} \right\} dx$

6. $\int \tan^3 x \sec^3 x \, dx = \int \tan^3 x \sec x \left(\sec^2 x \, dx\right)$ $u = \tan x$

$$\int \tan^2 x \sec^2 x \left(\sec x \tan x \, dx\right)$$

$\tan^2 x$ ← $\sec^2 x - 1$ ← $u^2 - 1$; $\sec^2 x$ → u^2; $(\sec x \tan x \, dx)$ → du

$$\boxed{\begin{aligned} u &= \sec x \\ du &= \sec x \tan x \, dx \end{aligned}}$$

$$\int (u^2-1)u^2 \, du = \int u^4 - u^2 \, du = \frac{1}{5}u^5 - \frac{1}{3}u^3 + C$$

$$= \frac{1}{5}\sec^5 x - \frac{1}{3}\sec^3 x + C$$

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16. $\int \frac{dx}{x\sqrt{9-x^2}}$

$a^2 - u^2$

$a = 3 \quad u = x$

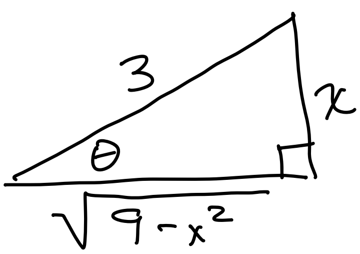
$$\boxed{\begin{aligned} x &= 3\sin\theta \\ dx &= 3\cos\theta \, d\theta \\ \sqrt{9 - 9\sin^2\theta} &= \sqrt{9(1-\sin^2\theta)} \\ &= 3\cos\theta \end{aligned}}$$

$$= \int \frac{3\cos\theta \, d\theta}{3\sin\theta \cdot 3\cos\theta} = \frac{1}{3}\int \frac{1}{\sin\theta} d\theta = \frac{1}{3}\int \csc\theta \, d\theta$$

$$= \frac{1}{3}\ln\left|\csc\theta - \cot\theta\right| + C$$

$\csc\theta$: $\frac{\text{hyp}}{\text{opp}}$ $\cot\theta$: $\frac{\text{adj}}{\text{opp}}$

$$= \boxed{\frac{1}{3}\ln\left|\frac{3}{x} - \frac{\sqrt{9-x^2}}{x}\right| + C}$$



$\sin\theta = \frac{x}{3}$

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2. $\int_1^3 x^2 \ln x\, dx = \frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^3 \cdot \frac{1}{x}\, dx$

$$\boxed{\begin{array}{ll} u = \ln x & dv = x^2\,dx \\ du = \frac{1}{x}\,dx & v = \frac{1}{3}x^3 \end{array}}$$

$= \frac{1}{3}x^3 \ln x - \frac{1}{3}\int x^2 dx$

$= \frac{1}{3}x^3 \ln x - \frac{1}{3} \cdot \frac{1}{3}x^3 \Big]_1^3$

$= \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 \Big]_1^3 = \left(\frac{1}{3}(27)\ln 3 - \frac{1}{9}(27)\right) - \left(\underbrace{\frac{1}{3}(1)\ln(1)}_{0} - \frac{1}{9}(1)\right)$

$= 9\ln 3 - 3 + \frac{1}{9}$

$= 9\ln 3 - \frac{26}{9}$

4. $\int \frac{x^2+1}{e^x}\, dx = \int (x^2+1)\, e^{-x}\, dx$ Tabular

$+$	x^2+1	e^{-x}
$-$	$2x$	$-e^{-x}$
$+$	2	e^{-x}
$-$	0	$-e^{-x}$

$$\boxed{(x^2+1)(-e^{-x}) - 2x(e^{-x}) - 2e^{-x} + C}$$

3. $\int_1^{e^2} \ln(x)\,dx \quad = x\ln(x) - \int x \cdot \frac{1}{x}\,dx$

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$$\boxed{\begin{array}{ll} u = \ln(x) & dv = dx \\ du = \frac{1}{x}\,dx & v = x \end{array}} = x\ln(x) - \int dx$$

$$= x\ln x - x \Big]_1^{e^2}$$

$$= \Big(e^2 \underbrace{\ln(e^2)}_{2} - e^2\Big) - \Big(1 \cdot \underbrace{\ln(1)}_{0} - 1\Big)$$

$$= 2e^2 - e^2 + 1 = \boxed{e^2 + 1}$$

7. $\int \sin^2 x \cos^5 x\,dx = \int \sin^2 x \, \underline{\cos^4 x} (\cos x\,dx)$ $\quad u = \sin x$, $du = \cos x$

$$= \int \sin^2 x \, (\underbrace{\cos^2 x}_{1-\sin^2 x})^2 \cos x\,dx$$

$$= \int u^2 (1-u^2)^2\,du = \ldots$$

17. $\displaystyle\int \frac{dx}{\sqrt{x^2-6x+13}}$

$(x^2-6x+\underline{9})+13-\underline{9}$

$(x-3)^2+4$

$$\int \frac{dx}{\sqrt{(x-3)^2+4}} = \int \frac{2\sec^2\theta\, d\theta}{2\sec\theta} = \int \sec\theta\, d\theta$$

u a

$x-3 = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

$\sqrt{(2\tan\theta)^2+4} = \sqrt{4\tan^2\theta+4}$

$= \sqrt{4(\tan^2\theta+1)} = \sqrt{4\sec^2\theta}$

$= 2\sec\theta$

$= \ln|\sec\theta+\tan\theta|+C$

$$\boxed{= \ln\left|\frac{\sqrt{(x-3)^2+4}}{2}+\frac{x-3}{2}\right|+C}$$

$\tan\theta = \dfrac{x-3}{2}$